The Augmented Synthetic Historical Control Method

Introduction

The synthetic control method (SCM) by Abadie et al. [2010] is one of the most popular methods in panel data econometrics for treatment effect estimation, second only to difference-in-differences. The idea of an SCM is very simple: it posits that the treated unit's untreated potential outcomes may be understood as some function/weighted average of a set of donor pool units, or unexposed/untreated entities. Many extensions have been proposed, including singular value decomposition [Amjad et al., 2018, Costa et al., 2023], penalized regression models [Abadie and L'Hour, 2021], Bayesian methods [Fernández-Morales et al., 2025, Kim et al., 2020], and other extensions [Abadie, 2021].

As with difference-in-differences, SCMs generally rely on the availability of a donor pool to compare the treated outcome to. This may not always hold in practice, especially with global events like pandemics or financial crises, or in instances where data for suitable donor units are impossible to collect. Chen et al. [2024] develop the synthetic historical control method (SHC). Instead of untreated donors, SHC leverages the pre-treatment time series of the treated unit as its own donor pool. The method compares a pre-treatment evaluation segment to other historical periods of the same unit, which are constructed by breaking the pre-treatment period into overlapping segments. The basic principle is that, all else being equal, future outcomes will resemble the past such that a weighted average of them may reconstruct the evaluation time series outcomes. Specifically, Chen et al. [2024] uses quadratic programming to solve for the weights of the respective donor segments, using a weighted average of them to predict the post-treatment counterfactual.

Chen et al. [2024], like Abadie et al. [2010, 2015] use convex weights to obtain in-sample fit. This means that the model's predictions are restricted only to the values of time segments that have been observed before, and may not be suited for complex time-series data where noise or non-smooth dynamics make a pure convex average inadequate. In such cases, exact (or even approximate) pre-treatment fit may be impossible, a challenge that also arises in standard SCM applications [Abadie et al., 2010].

This paper introduces the Augmented Synthetic Historical Control Method (ASHC), which extends SHC in two key ways. First, it relaxes the strict simplex constraint to an affine weight constraint, requiring only that the weights sum to one. This allows for negative weights, enabling controlled extrapolation beyond the convex hull of historical segments. Second, ASHC introduces a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM of Ben-Michael et al. [2021]. This balances flexibility with stability: extrapolation is possible, but penalized to avoid overfitting.

Importantly, ASHC retains the stabilizing Gram penalty $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2$ originally proposed by Chen et al. [2024], which ensures robustness in low-variance directions of the donor matrix. Together, these three components—the fit term, the SHC-proximity penalty, and the Gram penalty—define a new estimator that generalizes SHC while preserving its semiparametric time-series foundation.

From a theoretical standpoint, ASHC departs from the traditional linear factor model framework used to justify SCM. In SCM, identification is motivated by assuming that untreated potential outcomes follow a factor structure across units. By contrast, SHC and ASHC operate in a purely time-series context: the donor pool consists of overlapping pre-treatment segments of the treated unit itself. Our assumptions therefore follow the semiparametric regression tradition for time-series data. Specifically, we assume that the latent trend is smooth enough to be well-approximated by a truncated Taylor expansion, that noise is bounded (or sub-Gaussian), and that the donor matrix has a bounded operator norm. These conditions, while weaker than a full parametric data-generating process, are sufficient to establish finite-sample deviation bounds, asymptotic consistency, and convergence rates for the ASHC estimator.

A crucial aspect of the ASHC method is the choice of the regularization parameter, λ, which controls the trade-off between fitting the data and staying close to the SHC solution. The objective function for ASHC is given by:

$$\min_{w} \frac{1}{2\lambda} \sum_{t\in\text{eval}} (y_t - \hat{y}_{t|w})^2 + \|w - w_{\text{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2$$

where y_t is the outcome at time t, $\hat{y}_{t|w}$ is the counterfactual prediction, w are the ASHC weights, w_{SHC} are the original SHC weights, ς is the Gram penalty parameter, and $\mathbf{C}_0^\top$ is a matrix of eigenvectors corresponding to small eigenvalues of the donor matrix.

We propose a holdout validation approach to select the optimal λ. We split the pre-treatment period into a training and a validation set. For each candidate value of λ in a predefined grid, we solve the ASHC quadratic program using the training data to obtain a set of weights. These weights are then used to predict the outcomes for the validation data and the mean squared error (MSE) is calculated. The λ that minimizes the validation MSE is chosen as the optimal parameter. This procedure ensures that the chosen λ provides the best balance of fit and stability on unseen data within the evaluation period itself.

This paper makes two main contributions. First, we introduce the ASHC estimator, which extends the synthetic historical control framework of Chen et al. [2024] by permitting limited extrapolation beyond the convex hull of donor segments. The estimator combines three components: a fit term that aligns ASHC weights with the evaluation period, a proximity penalty that discourages large deviations from the SHC weights, and a stabilizing penalty that regularizes low-variance directions. Together, these features balance flexibility and robustness, allowing ASHC to provide feasible and stable solutions even in complex or noisy time-series settings.

Second, we establish a set of theoretical properties that clarify the performance guarantees of ASHC. We begin by deriving a finite-sample error bound, showing that ASHC prediction error decomposes into four interpretable components: the SHC residual fit error, a deviation term controlled by effective regularization strength, a smoothness-induced bias from truncating the Taylor approximation of the latent trend, and an additive noise term. We then prove that ASHC is asymptotically consistent: as the length of the pre-treatment period and the smoothing order increase, the prediction error converges to the noise floor, and under sub-Gaussian noise, it converges to zero in probability. Finally, we establish the convergence rate, demonstrating that under an optimally chosen bandwidth, the prediction error decays at rate $O_p(m^{-H/(2H+3)})$, approaching the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order grows. Together, these results show that ASHC preserves the interpretability of SHC while offering provably tighter error control and stronger asymptotic guarantees.

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